

MSc Project Report

School of Physics, Engineering and Computer Science.

University of Hertfordshire

DESIGN & ANALYSIS OF BODY WORK FOR AN AUTONOMOUS RACING VEHICLE

Report by

RUPESHWARAN PARANTHAMAN

Supervisor

ESFANDIAR KHALEGHI

Date

27/04/2026

PLAGIARISM STATEMENT

I have read the detailed guidelines to Students on academic offences and misconduct information (Refer to the Ask Herts web page for the [Assessment Offences and Academic Misconduct](#)).

I have read and understood the information posted on the Canvas Project module page about “Academic integrity, misconduct, plagiarism and unauthorised use of Artificial Intelligence (AI) in researching or presenting material for an assessment”. (Click this [Ask Herts link](#) for more information)

I have attended the mandatory seminar delivered by SASH and/or the School Academic Integrity Officer (SAIO) about Assessment Offences and Academic Misconduct.

I understand the University has zero tolerance for academic offences and misconduct, any suspected case may be subject to an investigation by the SAIO and disciplinary penalties may apply if the misconduct case is confirmed. (Reference: [University Policy Regulation AS14, Appendix III – version 17.0](#))

Hereby, I certify that the work submitted is my own and that any material derived from AI or quoted from any published or unpublished work of other persons has been duly acknowledged.)

Signature:



Student name: RUPESHWARAN PARANTHAMAN
SRN:23069976

UNIVERSITY OF HERTFORDSHIRE
SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

ABSTRACT

This report examines the design and analysis of a lightweight CFRP body structure for an FS-AI vehicle, due to the necessity of meeting the requirements of new safety regulations and ease of manufacture. Relevant literature on composite sandwich structure design and student-scale manufacturing processes influenced the approach used in this project. The process followed involved the application of an engineering design approach based on rules analysis, a Product Design Specification, concept generation, and evaluation. The design chosen uses CFRP sandwich structures, formed using vacuum-assisted resin infusion methods. The findings reveal that this design satisfies the requirements, providing a lightweight, modular design feasible for student manufacturing.

ACKNOWLEDGEMENTS

The author wishes to express his gratitude to Dr **ESFANDIAR KHALEGHI** who has been the supervisor for this project and for providing guidance, feedback, and support during this process. The author wishes to thank the Formula Student Autonomous (FS-AI) team for technical knowledge and design references along with the provision of relevant data for this project. The author would also like to extend his gratitude to the School of Physics, Engineering and Computer Science, University of Hertfordshire for the availability of facilities.

TABLE OF CONTENTS

PLAGIARISM STATEMENT	i
ABSTRACT	i
ACKNOWLEDGEMENTS	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	iv
GLOSSARY	v
1. INTRODUCTION	1
1.1 Project Background	2
1.2 Aim and Objectives	6
2. LITERATURE REVIEW	7
2.1 Composite Materials in Motorsport Applications	7
2.2 Sandwich Panel Structures	8
2.3 CFRP Manufacturing Methods for Student Projects	8
2.4 Numerical Analysis of CFRP Structures Using ANSYS	9
2.5 Impact and Crash Behaviour of CFRP Sandwich Panels	9
2.6 Research Gap and Project Contribution	9
3. METHODOLOGY	10
3.1 Methodological Overview	10
3.2 Product Design Specification (PDS)	10
3.3 Concept Development and Evaluation	13
3.4 Material Selection and Design	15
3.5 Testing Methodology	21
3.6 Manufacturing Considerations	21
4. RESULTS AND DISCUSSION	26
4.1 Integrated Results and Discussion	29
REFERENCES	31
BIBLIOGRAPHY	32
APPENDICES	33

LIST OF FIGURES

Figure 1.1 Formula Student race car	2
Figure 1.2 Formula Student Driverless vehicle	3
Figure 3.1 Concept sketches	13
Figure 3.2 CFRP sandwich panel	15
Figure 3.3 ANSYS ACP model of CFRP panel	15
Figure 3.4 FS-AI chassis	16
Figure 3.5 Bodywork panels	17
Figure 3.6 Bodywork panels	17
Figure 3.7 ACP model of body panels	19
Figure 3.8 ACP model of body panels	19
Figure 3.9 ACP model of body panels	20
Figure 3.10 Dry lay-up of CFRP reinforcement	22
Figure 3.11 Vacuum bagging process	23
Figure 3.12 Vacuum application and fibre compaction	24
Figure 3.13 Resin infusion and curing stage	25
Figure 4.1 Total deformation response	26
Figure 4.2 Three-point bending test configuration	26
Figure 4.3 Shear stress distribution	27
Figure 4.4 Localised shear stress near mid-span	27
Figure 4.5 Equivalent (von Mises) stress contours	28
Figure 4.6 Spanwise variation of equivalent stress	28

LIST OF TABLES

Table 1.1 Problem Statement – 5W's	4
Table 1.2 Problem Statement – PoV	5
Table 2.1 Composite Materials in Motorsport Applications	7
Table 2.2 Manufacturing Methods Comparison	8
Table 3.1 General Design Requirements	10
Table 3.2 Material and Structural Requirements	10
Table 3.3 Interface and Compatibility Requirements	11
Table 3.4 Material Requirements	11
Table 3.5 Operating Environment Requirements	11
Table 3.6 Performance Requirements	11
Table 3.7 Safety Requirements	11
Table 3.8 Aesthetics and User Experience	12
Table 3.9 Manufacturing and Cost Targets	12
Table 3.10 Sustainability Targets	12
Table 3.11 Summary of Proposed Bodywork Concepts	13
Table 3.12 Evaluation Criteria and Weightings	14
Table 3.13 Weighted & Ranked Scoring Matrix	14
Table 3.14 Final Concept Ranking	14
Table 3.15 CFRP Laminate Material Properties	18
Table 3.16 Estimated Bodywork Mass	19
Table 4.1 Summary of Numerical Three-Point Bending Results	30

GLOSSARY

Term	Definition
ACP	Composite PrepPost – ANSYS module used for composite laminate definition, lay-up modelling, and post-processing analysis.
ANSYS	Engineering simulation software used for finite element analysis of structures and materials.
CFRP	Carbon Fibre Reinforced Polymer – a lightweight composite material used for high-stiffness body panels.
FEA	Finite Element Analysis – numerical simulation technique used to evaluate stiffness, deformation, stress, and modal behaviour.
FS-AI	Formula Student Driverless – autonomous vehicle category of the Formula Student competition.
GPa	Gigapascal – unit of stress or elastic modulus commonly used in material property definitions.
PDS	Product Design Specification – document outlining all technical, functional, and regulatory requirements of the project.
VARI	Vacuum-Assisted Resin Infusion – composite manufacturing process used to produce CFRP sandwich panels.
Sandwich Panel	Composite structure consisting of two CFRP face sheets bonded to a lightweight core to achieve high bending stiffness.
Core Material	Lightweight structural layer (e.g. Nomex or PMI foam) placed between CFRP skins to increase stiffness and resist shear loads.
CAD	Computer-Aided Design – three-dimensional modelling process used for bodywork and chassis integration.
Edge Radius	Minimum curvature required by FS-AI regulations to ensure safe external bodywork edges.
Modularity	Ability to remove or replace bodywork panels easily during testing and maintenance.
Scrutineering	Technical inspection procedure at Formula Student events used to verify safety and regulatory compliance.

1. INTRODUCTION.

Formula Student Driverless (FS-AI) is a sophisticated engineering competition within motor racing where university-based teams construct autonomous racing cars. Contrary to other FS classes, no driver is present inside FS-AI cars, making safety and robustness even more important aspects. Hence, regulations require the bodywork of such a vehicle to satisfy strict constraints related to continuous exterior coverage, appropriate edge radii, removal of sharp edges, and protection of autonomous components. Besides the required properties, bodywork should be efficient from an aerodynamics standpoint while being lightweight and manufacturable at a student level. Composite materials made of carbon fiber reinforced polymers (CFRPs) are popular in motor sports because of their high stiffness-to-weight ratios and excellent mechanical properties. The composite sandwich panels, which consist of two CFRP sheets adhered to each other via a light core, can provide great bending stiffness with a negligible amount of weight, thus representing a good option for the bodywork design. The suitability of CFRP sandwich construction to Formula Student applications justifies its use for this purpose. Yet, the design and fabrication of compliant CFRP bodywork at a student-level manufacturing facility present certain challenges. The idea behind this project was prompted by the examination of the current bodywork of the team's Formula Student Autonomous car that failed to meet the most recent FS-AI regulations. The gaps observed included poor aerodynamic surface continuity, lack of safety coverage, poor edge finishing, and unfeasible manufacturing procedures. The failure to develop a coherent approach for the development of the bodywork meant that it is fragmented and manufactured inefficiently. These reasons made it clear that there was a necessity to create a new system to address all these problems and fit the current regulations.

This project aimed at developing a full CFRP bodywork system for a Formula Student Driverless car to be competed in the season of 2026. In this regard, an engineering approach was applied throughout this project that started with a thorough investigation of the current regulations of the event along with research into composite materials and bodyworks in particular. An elaborate Product Design Specification (PDS) was drafted to determine the requirements for design and turn them into measurable goals. Several different bodywork designs were developed, analyzed and compared by using a weighted decision matrix. This particular concept was further designed in elaborate 3D CAD models specifying the shape of panels, splitting lines, mount points, and surface continuity. A feasible manufacturing approach using Vacuum Assisted Resin Infusion (VARI) technology was suggested, which is consistent with the manufacturing processes adopted

DESIGN & ANALYSIS OF BODY WORK FOR AN AUTONOMOUS RACING VEHICLE

by most FS teams. The report ends with an evaluation of the efficiency of the design proposal along with areas for further investigation in the future.

1.1 *Project Background.*

Formula Student Driverless (FS-AI) is an extremely challenging engineering design programme wherein university teams build completely autonomous race cars. Unlike traditional Formula Student class cars, FS-AI vehicles do not require a human operator and place greater attention on safety, reliability, and regulatory conformity. As such, bodywork design for FS-AI vehicles assumes vital importance in guaranteeing continual external coverage, conforming edge curves, sensor protection, and predictable airfoil characteristics.



Figure 1.1 Formula student Car (source: Formula Student (2025)). Image showing a typical Formula Student vehicle used in university-level motorsport competitions. Accessed: 21 April 2026.

There have been numerous recent regulations governing the development of Formula Student race cars, especially regarding bodywork shape, continuity, and safety aspects. Such changes have further complicated the design process for students, demanding more systematic and engineering-based approaches to bodywork design. Composites are commonly used materials in motor sports because of their outstanding stiffness-to-weight ratio and smooth aerodynamic surfaces. Particularly, carbon fiber reinforced polymer (CFRP) sandwich materials constitute light and stiff bodywork panels, which makes them ideal materials for Formula Students.

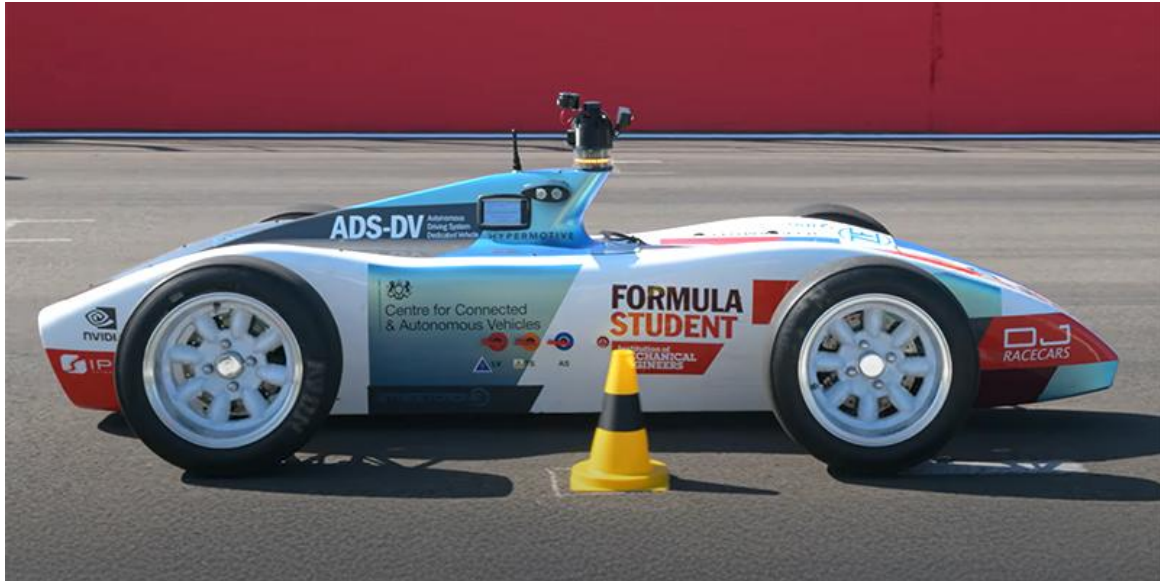


Figure 1.2 Formula Student Driverless (FS-AI) vehicle (source: Formula Student (2025)). Illustration of an autonomous Formula Student vehicle highlighting the bodywork coverage and sensor placement. Accessed: 21 April 2026.

Evaluating the bodywork of the FS-AI vehicle, various weaknesses were found, namely, lack of compliance with new standards, poor surface transitions, and difficulty in manufacture through student means. There is, therefore, a need for an all-round overhaul of the bodywork system to ensure that it meets the required regulatory and aerodynamic standards and can be easily manufactured by students. It was against this background that the current research was conceived.

A review of the present bodywork on the team's Formula Student Driverless (FS-AI) race car revealed some issues in comparison to the latest set of regulations. The current bodywork design failed to meet the criteria regarding constant outer cover, acceptable radius edges, and sufficient autonomous equipment protection. Additionally, the shape and manufacture of existing panels did not follow any common design approach, leading to the use of multiple elements that were difficult to produce using student-level composite techniques.

The aforementioned problems might have affected the car's eligibility due to non-conformity during scrutineering, and manufacturing, assembling, and servicing would have been inefficient. Given the new FS-AI regulations for the 2026 racing season, there was a need for a well-developed and organized approach to the bodywork design. This explains why the proposed project aimed at designing the entire bodywork system, which had to comply with the regulations, be lightweight and durable, and manufacturable using student-level composite technology.

Element	Description
Who	Formula Student Driverless (FS-AI) vehicle designers, manufacturers, and scrutineers responsible for ensuring regulatory compliance, safety, and manufacturability of the vehicle bodywork.
What	The existing bodywork design does not fully comply with the latest FS-AI regulations and lacks a unified, lightweight, and manufacturable CFRP bodywork solution. Issues include inadequate surface continuity, inconsistent edge geometry, and limited compatibility with student-level composite manufacturing processes.
Where	The problem is specific to the external CFRP bodywork panels mounted on the Formula Student Driverless chassis, including the nose section, side panels, and top body enclosure.
When	During preparation for the FS-AI 2026 competition season, following the introduction of updated bodywork and safety regulations requiring stricter compliance during scrutineering.
Why	Non-compliant or poorly designed bodywork increases the risk of failing scrutineering, introduces unnecessary mass and manufacturing complexity, and limits structural validation and repeatability. A redesigned solution is required to ensure compliance, safety, and manufacturability.

TABLE 1.1 PROBLEM STATEMENT 5 W'S.

The use of the 5W technique ensured that the problem tackled in this study was well-defined by asking questions about who was affected by the problem, what the problem entailed, where it occurred, when it became crucial, and why it needed to be solved. With the help of this structured approach towards defining the problem, the technique ensured an informative description of the design problems posed by the current FS-AI bodywork.

User / Stakeholder	User Need	Design Insight (Engineering PoV)
FS-AI Vehicle Design Team	A bodywork system that complies with FS-AI regulations while remaining lightweight and manufacturable within limited student resources.	Without a structured CFRP bodywork design strategy, teams risk non-compliance during scrutineering and excessive re-work. A unified CFRP sandwich panel concept with defined geometry, thickness, and mounting interfaces reduces risk and improves design repeatability.
Manufacturing Team	CFRP panels that can be produced reliably using student-accessible composite processes with minimal defects.	Complex geometries and inconsistent laminate designs increase voids, scrap rate, and manufacturing time. Flat or singly-curved CFRP sandwich panels designed specifically for VARI improve laminate quality and production consistency.
Scrutineering Officials	Bodywork with continuous coverage, compliant edge radii, and safe external geometry.	Late identification of sharp edges or incomplete coverage results in last-minute modifications. Integrating FS-AI bodywork rules directly into the CAD and composite design process ensures first-time compliance during inspection.
Structural / Analysis Engineer	A bodywork design that can be validated using numerical simulation and physical testing.	Inconsistent material definitions prevent accurate structural validation. Defining a standardised 2 mm CFRP laminate and validating it through ANSYS ACP and three-point bending tests enables reliable correlation between simulation and experimental results.
Vehicle Assembly & Maintenance Team	Modular bodywork panels that can be removed and reinstalled quickly during testing and competition.	Non-modular body panels slow down maintenance and risk damage during removal. A modular CFRP panel layout aligned with chassis hard points improves serviceability without compromising structural integrity.

TABLE 1.2 PROBLEM STATEMENT POV.

PoV methodology was employed in articulating the problem statement with respect to the stakeholders' perspectives for the Formula Student Driverless initiative. With PoV, through the identification of the needs of the users and their corresponding design insights, it becomes clear how the engineering problems associated with body work design, materials choice, testing process, and manufacturing play a critical role. Such an analysis is essential

in making sure that the solution proposed by the use of CFRP body work takes into account all the necessary considerations for its implementation.

1.2 *Aim and Objective.*

The objective of this work is to design, analyse and assess the structure of a lightweight and standard compliant CFRP sandwich panel bodywork assembly for a FS-AI formula student driverless race car with special consideration to the structural stiffness, manufacturability at student-level composite technology and conformity to bodywork safety standards for the FS-AI. The project is concerned with verifying the structural performance of the designed bodywork by means of composite modelling and three-point bending analysis.

In order to meet the project objective, the following sub-objectives will be addressed:

1. Literature research and motorsport practice review with respect to the following subjects: composites of CFRP type; sandwich panels; bodywork design for formula cars (Formula Student) application.
2. Analysis of current FS-AI bodywork rules and development of design constraints for safety and geometrical issues including edge radius and outer covering criteria.
3. Preparation of the Product Design Specification (PDS) by translating regulations and other factors into quantitative design criteria.
4. Generation of different bodywork design ideas based on the PDS, as well as selection of the optimal one according to the weighted criteria evaluation.
5. Designing of modular CFRP sandwich bodywork based on 3D CAD modeling and providing of the accurate fit to FS-AI chassis assembly.
6. Laminate architecture definition and modeling using ANSYS Composite PrepPost (ACP) in view of anisotropy of the material at the ply level.
7. Assessment of bodywork structural characteristics by means of three-point bending simulation.

2. LITERATURE REVIEW.

Formula Student is an international engineering competition that offers participants the ability to engineer, construct, and test fast race cars[1]. Formula Student Driverless (FS-AI) adds new challenges that demand the autonomy of vehicles, hence introducing new requirements for the safety of such vehicles. Under these conditions, bodywork performs several roles, such as ensuring safety, having round edges, safeguarding autonomous systems, and ensuring predictability during driving[1].

Current FS-AI regulations have placed special attention on completely enclosable bodywork, minimum acceptable edge radii, minimal protrusion from the chassis, and mounting security[1]. The above factors affect the choice of materials used, geometry of panels, and even manufacturing methods. As such, bodywork no longer acts as a purely aesthetic or aerodynamic element[1] of a vehicle but a vital safety structure that should be able to handle loads and other stresses applied to it. Areas were considered and the reasons why they were not utilised in the project should also be included.

2.1 Composite Materials in Motorsport Applications.

CFRPs have been applied extensively in motorsport owing to their better mechanical properties than metals. They are lightweight, flexible, and capable of forming intricate shapes – making CFRPs ideal for construction of racecar body components[2]. It becomes possible to make strong but lightweight panels, with the added advantage of having sleek exteriors[2].

The use of CFRPs in racing vehicles and cars generally is mainly for body panels and protective enclosures, as well as aerodynamic equipment[3]. Research findings indicate that CFRPs offer better specific stiffness and strength as compared to aluminium or steel counterparts. This translates into weight reduction without any degradation in structural rigidity[3]. On the other hand, performance characteristics of CFRPs vary considerably depending on orientation of fibres and laminate stack-up[4].

Application Context	Material System	Structural Form	Reported Advantages	Source
Formula Student vehicles	CFRP	Sandwich panels (skins + foam core)	High stiffness-to-weight ratio, student manufacture	Literature
Motorsport bodywork	CFRP	Thin monocoque / panel skins	Lightweight, smooth surface finish	Literature
Low-budget racing	GFRP / composites	Solid laminates	Low cost, reduced performance	Literature

Table 2.1 Composite Materials in Motorsport Applications

2.2 Sandwich Panel Structures.

A sandwich panel design is a proven method that has been used extensively in structural engineering to achieve high efficiency through lightness[4]. The basic sandwich panel is made up of two stiff CFRP sheets with a lightweight core material in between, like Nomex honeycomb or polymers foams[4]. The resulting sandwich panel design allows for increased bending rigidity with very little weight increase, which makes it an effective choice for body panels undergoing bending and handling stresses[4]. Research indicates that sandwich constructions have very good strength characteristics, excellent vibration qualities, and high damage resistance properties if well designed. In Formula Student vehicles, it is common practice to use sandwich panels for body panels, floor panels, and aerodynamic panels, owing to their high stiffness-to-weight ratio[5]. However, wrong choice of the core materials, poor edge sealing, and weak bonding could lead to early failure[5].

2.3 CFRP Manufacturing Methods for Student Projects.

A number of manufacturing techniques exist for fabricating CFRP parts; these include hand lay-up, vacuum bag molding, autoclave molding, and vacuum-assisted resin infusion (VARI)[6]. From all these methods, the VARI process is commonly used by Formula Student teams as it offers a relatively cheaper tooling cost, high-quality laminate[6], and capability of making large panels. In the VARI technique, dry fibers are first placed in a mold, vacuum pressure applied, and then resins infused into the laminate.

Studies reveal that the VARI method is capable of producing laminate with reduced voids and fiber-resin ratio when parameters are adequately controlled[7]. It is an ideal choice for student teams since it strikes the right balance between performance, efficiency, and cost-effectiveness[7]. Nonetheless, the VARI process entails proper design of flow paths, viscosity of resins, and vacuum sealing in order to prevent defects like dry spots.

Manufacturing Method	Typical Use Case	Key Characteristics	Relevance to FS-AI
Vacuum-Assisted Resin Infusion (VARI)	Formula Student teams	Low tooling cost, good laminate quality	Highly suitable
Out-of-Autoclave Prepreg	Motorsport / aerospace	Higher quality, higher cost	Limited by resources
Hand lay-up	Entry-level composites	Low cost, variable quality	Less suitable

Table 2.3 CFRP Manufacturing Methods for Student Projects.

2.4 Numerical Analysis of CFRP Structures Using ANSYS.

FEA plays an essential role in designing and analyzing[8] composite structures. Programs like ANSYS help create models for CFRP by specifying the material properties (orthotropy), ply angles, and stacking sequences. Unlike isotropic materials, there is more work involved in analyzing composites because of their nature[8].

ANSYS offers several features which make composite analysis possible, including creating a layered shell model, calculating laminate stiffness, and assessing stress conditions. Numerous studies show that the use of numerical analysis can accurately determine deformation behavior, stiffness behavior, and stress behavior for CFRP sandwiches. The ability to analyze materials before physically building[4] them makes simulation especially useful for student projects[8].

2.5 Impact and Crash Behaviour of CFRP Sandwich Panels.

Impact and crash response properties of CFRP structures are important parameters in applications where safety requirements apply[5]. Unlike metallic structures, CFRP composites have progressive failure behavior, which may involve matrix cracking, fibre breaking, and delamination. Sandwich structures, on the other hand, can fail through core crushing, as well as face-sheet delamination during an impact situation[5].

From the literature, it can be seen that low-speed impact and crash responses of composite structures can be successfully simulated using ANSYS explicit solvers[8]. Often, failure criteria and damage behavior of the material are calibrated by using experimental data obtained from material testing. In relation to FS body structure application, simulations involving low speed impact and loading situations are more important than crash tests[1].

2.6 Research Gap and Project Contribution.

Although there is abundant literature on CFRP, sandwich panels, and composite manufacturing techniques, there are relatively few sources that look into the application of compliant composite body work within the range of what is achievable by students[6]. Most of the publications concentrate on aerospace-grade composites or very expensive manufacturing techniques that do not apply to Formula Student competition[7].

The current project fills this research gap and explores regulations compliance and sandwich panel design in conjunction with structural and crash simulation using ANSYS while taking into consideration the limitation in manufacturing capability of student teams[1].

3. METHODOLOGY.

3.1 Methodological Overview

A structured approach was used throughout this project for the design, assessment, and validation of the CFRP bodywork system of a Formula Student Driverless car (FS-AI). The approach considered factors such as regulatory assessment, identification of design requirements, conceptual design, conceptual selection process, and manufacturing aspects to guarantee compliance, lightness, and feasibility of the design to manufacture by students.

The design process started from the formulation of a Product Design Specification (PDS) based on the FS-AI regulations, followed by design and evaluation of bodywork concepts until a final design concept selection. The methodological approach ensured consistency and repeatability of the design process before material modeling and structure assessment phases.

3.2 Product Design Specification (PDS)

The design specifications for the bodywork were obtained through an analysis of the existing FS-AI guidelines, operational conditions, and limitations related to the accessibility of composite construction by students. The design specifications were then compiled into a Product Design Specification (PDS). This document outlined the objectives of the bodywork design with respect to functionality, materials, structure, safety, and manufacturing processes. The PDS served as a benchmark during the design and evaluation process.

Requirement Category	Specification
Bodywork type	CFRP sandwich panel system
Competition compliance	FS-AI 2026 bodywork regulations
Operating environment	Outdoor track use (rain, debris)
Service life	Single FS-AI season

Table 3.1 – General Design Requirements.

Definition	Unit	Target	Comment
Overall envelope	mm	Within FS-AI allowed bodywork envelope	Clearance respected, no unsafe protrusions
Panel thickness	mm	2–6	Face sheets + core depending on stiffness
Core thickness	mm	5–8	PMI foam typical for student manufacture
Total mass	kg	≤ 10	Team target for performance optimisation

Table 3.2 – Material and Structural Requirements.

Definition	Unit	Target	Comment
Mount hole pattern	mm	Standard grid (20–25 spacing)	Consistent drilling, protected laminate
Fasteners	–	M6 countersunk with bonded inserts	Student-friendly assembly
Panel removal time	min	< 5	Serviceability during testing
Panel gap	mm	≤ 1–2	Helps maintain smooth external surfaces

Table 3.3 Interface and compatibility Requirements.

Definition	Unit	Target	Comment
Face-sheet fabric	gsm	200–300 (2×2 twill)	Common CFRP shell fabric
Core material	mm	5–8 Nomex / Pvc foam	High stiffness-to-weight
Resin system	–	Epoxy (VARI-compatible viscosity)	Suitable for infusion
Edge sealing	–	Required	Prevents moisture ingress and damage

Table 3.4 Material Requirements.

Definition	Unit	Target	Comment
Temperature	°C	0–60	Typical UK track conditions
Moisture	–	Rain / humidity	Structural integrity maintained when wet
UV resistance	–	Required	Clear-coat protection

Table 3.5 Operating Environment Requirements.

Definition	Unit	Target	Comment
Natural frequency	Hz	> 60–80	Avoids panel flutter near mounts
Factor of safety	–	≥ 2	Lifting and handling margin
Edge radius	mm	≥ 3–5	Meets FS-AI sharp-edge regulations
Surface continuity	–	Smooth, sealed	Inspection-friendly and stable

Table 3.6 Performance Requirements.

Definition	Unit	Target	Comment
Sharp edges	–	None permitted	Chamfered to comply with rules
Protrusions	–	Non-hazardous	No snag points
Fire / heat resistance	–	Epoxy CFRP norms	Isolated from hot components

Table 3.7 Safety Requirements.

Definition	Unit	Target	Comment
External finish	–	Smooth clear-coated or painted	Professional composite appearance
Colour	–	Team colours	Branding requirements
Installation time	min	< 5	Rapid panel changes during test days

Table 3.8 Aesthetics and User Experience.

Definition	Unit	Target	Comment
Manufacturing process	–	VARI (Vacuum-Assisted Resin Infusion)	Low tooling cost, reliable
Tooling	–	MDF / PU foam (sealed)	Reusable and low-cost
Scrap rate	%	< 10	Controlled ply-kit cutting
Repeatability	–	High	Standardised manufacturing schedule

Table 3.9 Manufacturing and cost Targets.

Definition	Unit	Target	Comment
Off-cut reduction	%	> 20	Nested ply kits
Rework	%	< 5	Improved vacuum bagging
Safe disposal	–	Team policy	Compliant waste handling

Table 3.10 Sustainability Targets.

3.3 Concept Development and Evaluation.

Several designs of bodyworks were created as ways to tackle various approaches to geometry, modularity, and manufacturability. Every design was analyzed in qualitative terms to make sure it fulfills basic requirements according to FS-AI before being subjected to further quantitative analysis. In turn, a Weighted and Ranked (W&R) decision matrix was used to objectively evaluate the designs based on criteria determined by PDS, such as manufacturability, simplicity, robustness, weight optimization, serviceability, and cost. The design with the best weighted score was selected as the basis for the final solution.



Figure 3.1 Concept sketches.

Concept definitions:

Concept 1 – Flat Top Minimal: Planar design focused on ease of production.

Concept 2 – Smooth Curved Aero: Fully curved enclosure with increased difficulty of production.

Concept 3 – Box Sides with Aero Top: Side enclosures with an aerodynamic top enclosure.

Concept	Key Features	Advantages	Limitations
Concept 1	Flat, simple panels	Easy manufacture, low mass	Reduced geometric refinement
Concept 2	Fully curved shell	Continuous geometry	Complex tooling
Concept 3	Modular sides + top	Balanced design	More panel seams

Table 3.11 – Summary of Proposed Bodywork Concepts.

Criterion	Weight (%)
Manufacturability	25
Structural simplicity	20
Compliance robustness	20
Mass efficiency	15
Serviceability	10
Cost	10
Total	100

Table 3.12 – Evaluation Criteria and Weightings.

Criterion	Weight	Concept 1	Concept 2	Concept 3
Manufacturability	25	5	2	3
Structural simplicity	20	4	3	3
Compliance robustness	20	4	4	4
Mass efficiency	15	5	3	3
Serviceability	10	4	3	4
Cost	10	5	2	3

Table 3.13 – Weighted & Ranked Scoring Matrix.

Concept	Total Score	Rank
Concept 1 – Flat-Top Minimal	73	1st
Concept 2 – Smooth Curved	65	2nd
Concept 3 – Box-Side + Aero Top	65	2nd

Table 3.14 – Final Concept Ranking

Justification for the Choice of the Selected Design Concept.

The first design concept earned the highest total score among all three designs during the Weighted and Ranked evaluation method because of its excellent performance in the most highly weighted criteria. The use of a simpler geometric form of the panels enhanced the manufactureability and structure simpleness and lowered the cost of manufacturing, considering that these factors are the priority for student-level manufacturing. Additionally, the first concept outperformed the others in terms of mass efficiency and ease of installation without any significant differences in terms of regulatory compliance. However, the second design concept was less favourable due to higher manufacturing complexity caused by fully curved panels, and the third design concept had panel joints, compromising structure simpleness.

3.4 Material Selection and Design.

With the selection of concepts being complete, detailed design of the bodywork was undertaken through 3D CAD models to allow accurate fitting on the vehicle chassis. Reference was made to the geometry of the chassis when defining mounting interfaces, clearance areas, and panel break lines. Modularity of the panels was designed into the bodywork to aid the ease of installation, removal, and maintenance for testing and racing purposes.

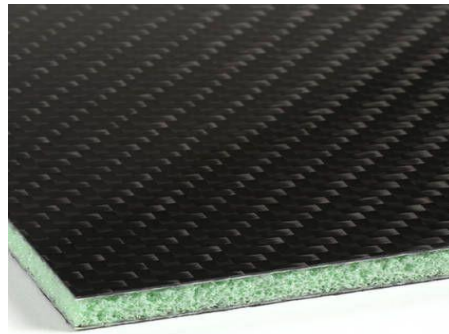


Figure 3.2 CFRP sandwich panel.

Selection of materials involved using carbon fibre reinforced polymer (CFRP) sandwich panels, owing to their high stiffness-to-weight ratios as well as their capability of being manufactured by student-level engineers.

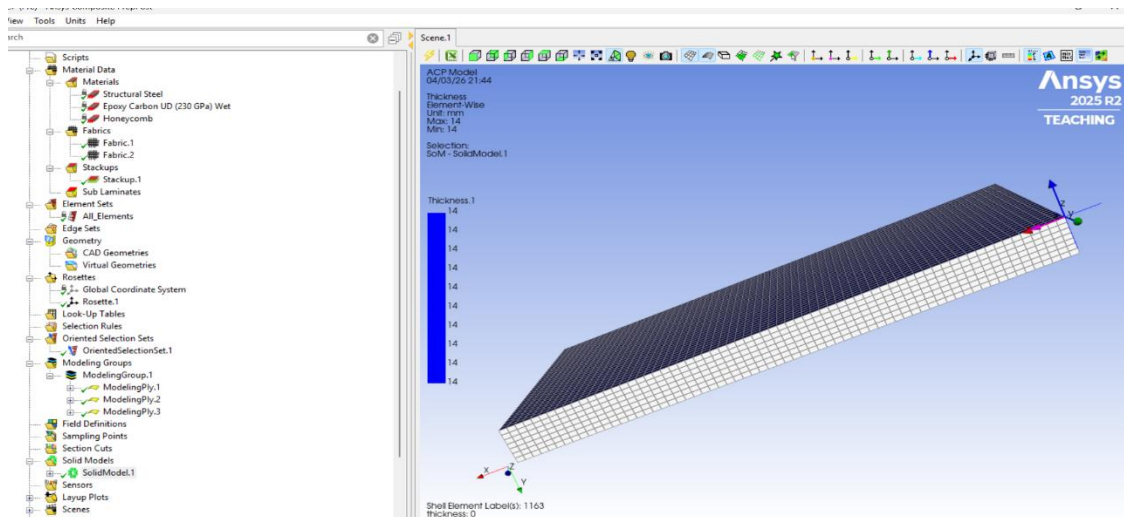


Figure 3.3 ANSYS ACP model of CFRP panel.

Design of composite laminates was done through the use of ANSYS Composite Prep Post (ACP), in order to accurately capture the anisotropic nature of CFRP materials. Material properties at the ply level were defined using carbon fibre fabric systems and epoxy resin matrix systems suitable for Vacuum-Assisted Resin Infusion (VARI). Stacking arrangements, fibre angles, ply thicknesses, and sandwich core designs were directly applied to the CAD geometry of the bodywork panels.

The use of ANSYS Composite Prep Post (ACP) assured consistency of the manufacturing process in the model prior to analysis.

After deciding on the final bodywork design, the next stage involved designing the bodywork and incorporating it with the Formula Student Driverless chassis. The chassis was made as a tubular space frame and acted as the main benchmark for the bodywork. This meant that some limitations in terms of geometry, mounting, and clearance would need to be considered while designing the bodywork. For this reason, a fully modeled three-dimensional CAD model of the chassis was used as the base for developing the bodywork.

As a space frame, the chassis indicated the packaging volume, load-bearing structures, and structural anchor points that the bodywork will have to consider. Special consideration was given to areas around the roll hoops, nose section, and side members, since these were the areas that would define panel edges and mounting locations. It is also worth noting that the bodywork needed to cover the entire surface of the chassis while satisfying the FS-AI safety requirements, such as minimum radius of curvature and absence of sharp edges.

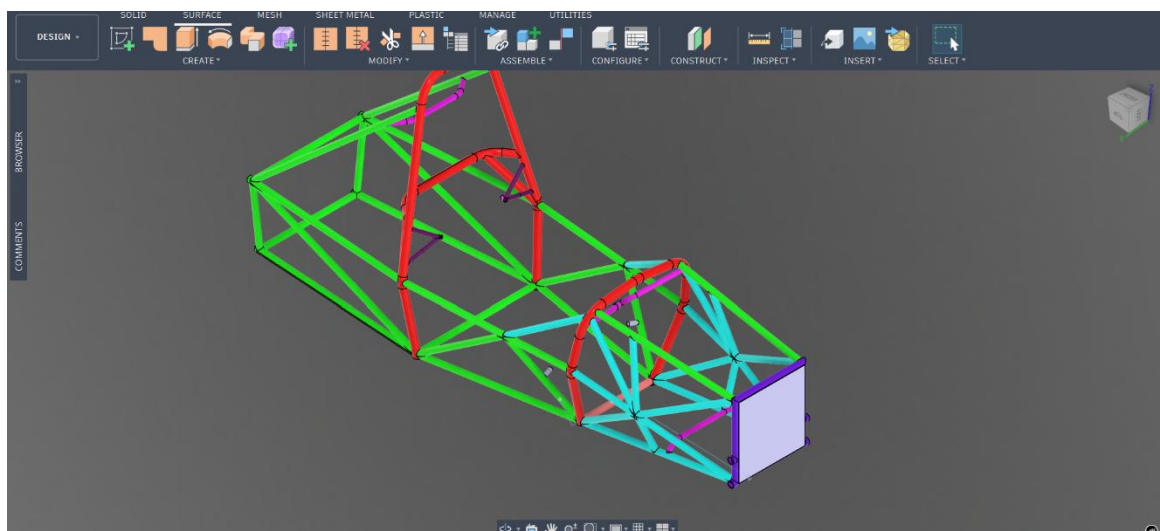


Figure 3.4 FSAI chassis.

Bodywork panels were developed based on the geometry of the chassis to facilitate precise assembly and fitment. Bodywork panels were developed in a modular manner to allow for easy assembly, dismantling, and servicing when needed during tests and races. The body panel split lines were placed in such a way as to coincide with structurally stable points of the chassis so as to keep local stresses low and to allow for reliable fixation.

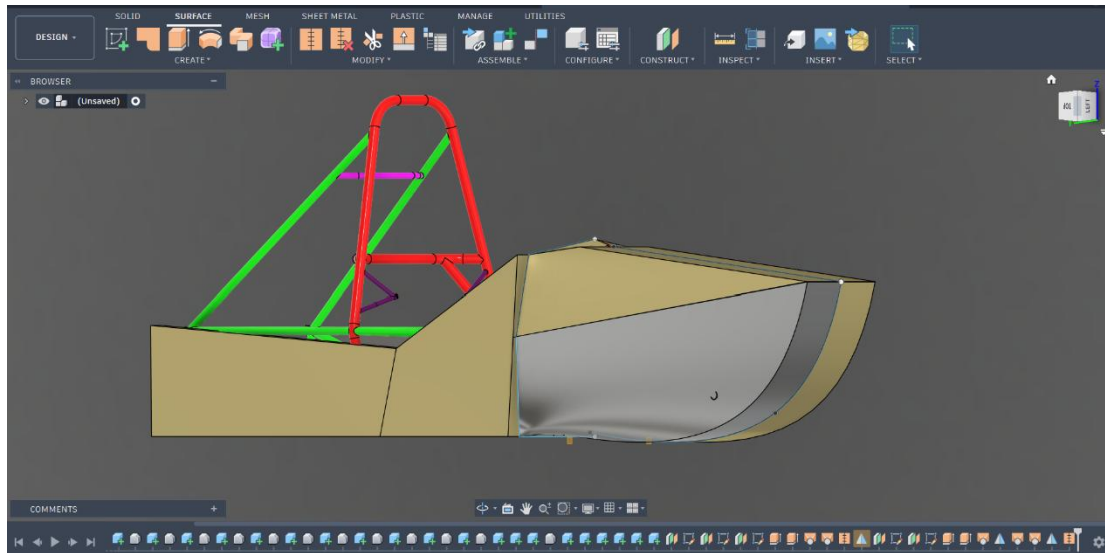


Figure 3.5 Bodywork panels.

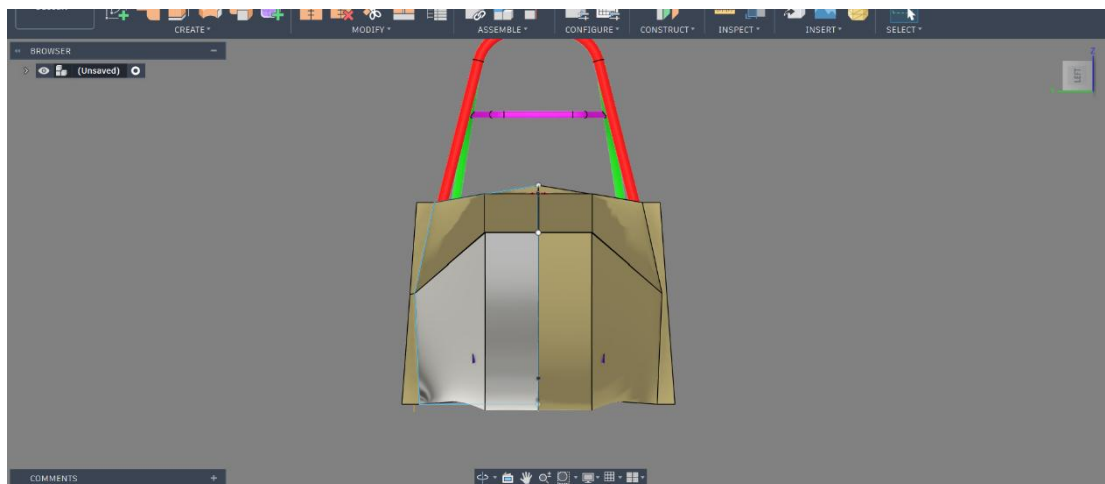


Figure 3.6 Bodywork panels.

The bodywork panels had been modeled as composite sandwich structures made from carbon fiber reinforced polymer (CFRP), as the material would provide both reduced weight and good stiffness/handling characteristics. In the context of material and numerical analyses as well as manufacturing processes, the CFRP laminates were considered as 2 mm thick. This value was chosen due to its applicability in manufacturing thin bodywork panels, common for use in Formula Student projects.

The choice of 2 mm thick CFRP laminate resulted from the need to achieve both good rigidity and low weight in the material. With such thickness, it would be possible to manufacture the panel by means of Vacuum-Assisted Resin Infusion (VARI), which can only be used for the manufacturing of thin panels. Furthermore, 2 mm laminate thickness allowed for numerical analysis through the use of ANSYS program, which facilitated direct comparison of experimental tests and simulations.

The sandwich structure principle involved using CFRP surface layers that were joined to a core material, which was relatively light. The function of the CFRP layers was to resist tensile and compressive loads along the plane of bending; the core would increase the section modulus and resist shearing forces. The sandwich principle provided considerable stiffness in bending for a comparable mass compared to an isotropic laminate, hence its suitability for large flat and singly curved body panels.

Consistent thickness and material layout throughout all the body panels simplified production, testing, and analysis. Areas where there were high local stresses, such as attachments and edge areas, would accommodate localized reinforcement with extra plies and inserts without interfering with the overall design of the panels.

Property	Symbol	Value	Unit
Laminate thickness	t	2.0	mm
Fibre material	—	Carbon fibre (woven)	—
Matrix material	—	Epoxy resin	—
Young's modulus ($0^\circ/90^\circ$)	E_1, E_2	50–70	GPa
In-plane shear modulus	G_{12}	4–6	GPa
Poisson's ratio	ν_{12}	0.05–0.1	—
Flexural stiffness (qualitative)	—	High	—
Manufacturing process	—	VARI	—

Table 3.15 CFRP property

Item	Approximate Mass
CFRP bodywork panels (2 mm sandwich)	5–6 kg
Bare chassis	Not assumed
Mass increase due to bodywork	+5–6 kg
PDS target	≤ 10 kg

Table 3.16 Total mass of the chassis.

Composite Laminate Definition Using ANSYS Composite Prepost(ACP).

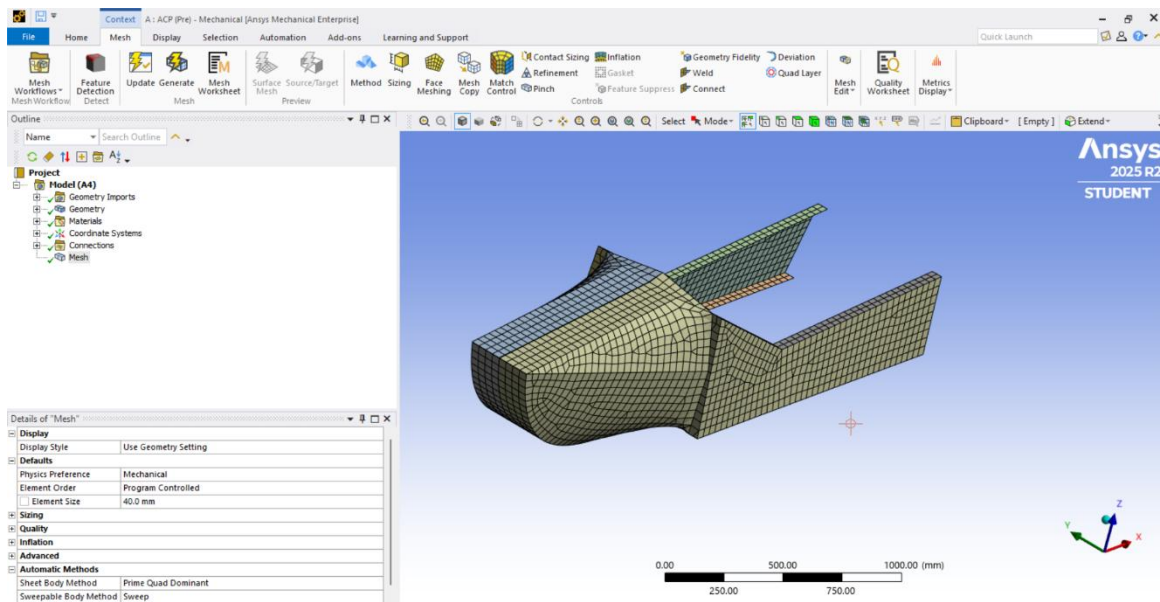


Figure 3.7 ACP model of Body panels

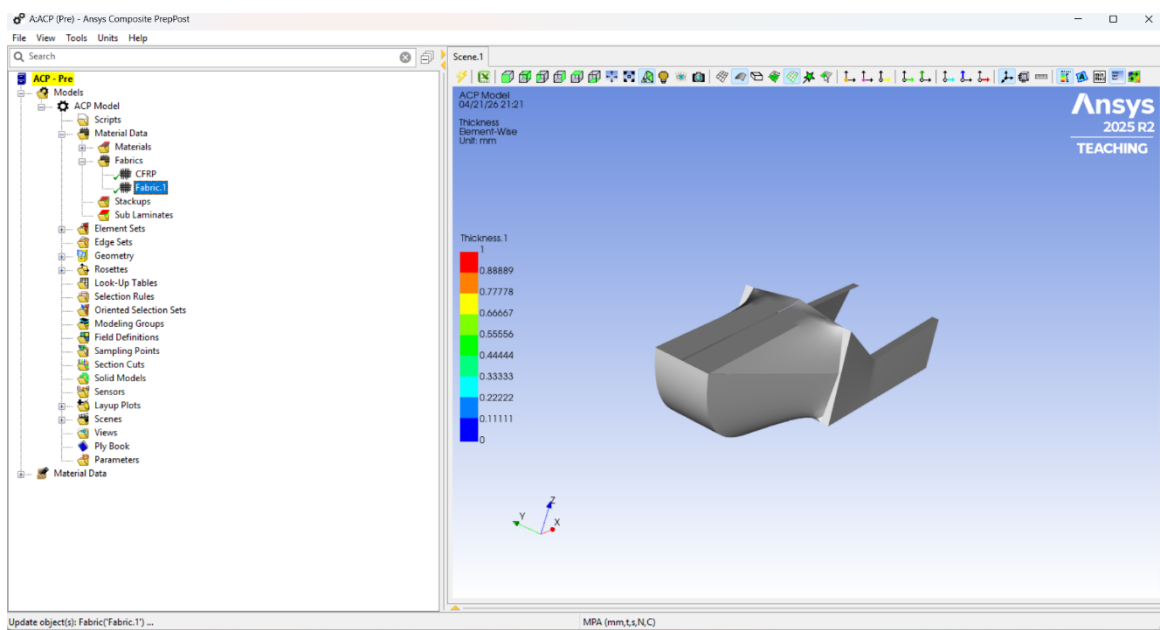


Figure 3.8 ACP model of Body panels.

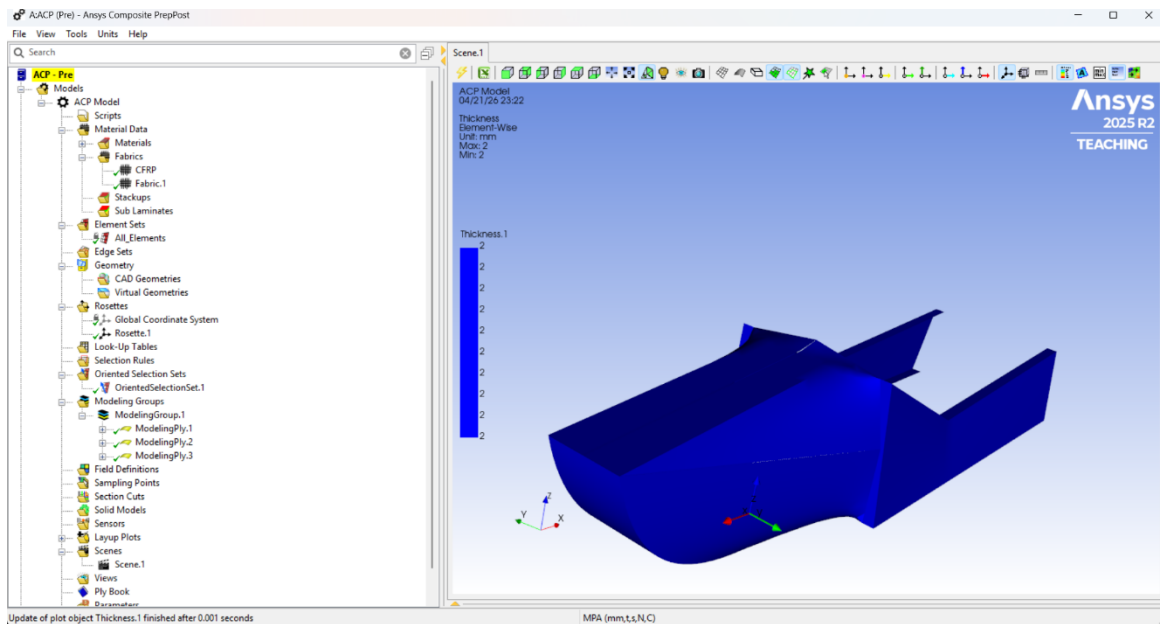


Figure 3.9 ACP model of the body panels.

The CFRP body panel model is shown in Figure 3.X and created via ANSYS Composite PrepPost (ACP). In this study, the body part has been assumed to be a sandwich composite component in order to reflect the actual nature of manufacturing and performance. As mentioned before, the laminate comprises a CFRP face sheet placed on the outer side and another one placed on the inner side with a foam core layer between them. This type of arrangement has been adopted to ensure maximum bending stiffness at minimum mass level, which is of high importance for Formula Student body components. The overall thickness of the composite laminate has been determined to be 2 mm, which is typical of lightweight body parts and corresponds to the specimens utilized for both numerical and experimental tests (three-point bending). In the case of ACP analysis, ply-level material properties, laminate lay-up, and foam core have been assigned to the CAD model in order to reflect the anisotropy of CFRP laminates and the shear capability of the foam core material.

3.5 Testing Methodology.

The testing of the CFRP panels for the bodywork was performed through a combination of numerical simulation and physical experimentation. The three-point bending test was carried out through numerical simulation in ANSYS for the flexural analysis and determination of the bending stiffness of the CFRP sandwich panels. Through the numerical simulation, it was possible to study the behavior of the laminate and determine the points where there is stress concentration as well as deformations.

Moreover, physical three-point bending was also carried out in laboratory conditions for representative samples of the CFRP sandwich structures that were produced using the same materials and stacking sequence as that designed at the concept stage. Physical experimentation was carried out under conditions similar to those used in the numerical models. The results from physical experimentation were compared to those from the numerical simulation in order to verify the correctness of the simulation results.

3.6 Manufacturing Considerations.

One of the factors that had to be taken into consideration during both the design and validation phases of this project is manufacturability. The composite laminate structures that were to be numerically analyzed through ANSYS Composite PrepPost (ACP) software were specifically chosen such that they can be manufactured using the Vacuum-Assisted Resin Infusion (VARI) technique. VARI is a composite manufacturing process that is widely used by Formula Student teams owing to its low-cost nature and reliable results.

Vacuum-Assisted Resin Infusion is a closed-mould composite manufacturing process where the fiber reinforcements and cores are first positioned inside a mold, and then they are infused with the resin under vacuum pressure. This process differs from the traditional wet-layup process in that the positioning of fibers and resin infusion take place separately, making the process more controllable and predictable.

Stage1: Dry lay up of CFRP Reinforcement.

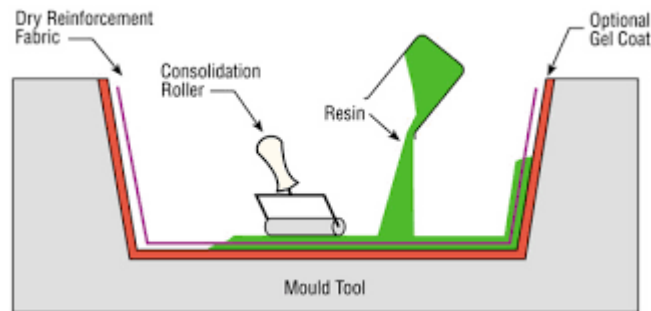


Figure 3.10: Dry lay-up of CFRP reinforcement (source: Explore Composites (2022)). Dry carbon fibre fabrics and core materials positioned inside the mould prior to resin infusion. Accessed: 21 April 2026.

The VARI manufacturing method starts with the dry lay-up process where dry carbon fibre reinforcement materials and core materials are inserted into the pre-formed mould without the resin. The carbon fibre reinforcement materials are first cut in accordance with the designed ply patterns and then oriented according to the required fibre orientations. In the case of CFRP sandwich panels, core material is placed between the two faces. The proper orientation of the fibres and core material is very important in order to determine the stiffness and bend properties of the laminate.

Stage 2: Vacuum bagging and consumable Placement.



Figure 3.11: Vacuum bagging process for composite laminates (source: Explore Composites (2022)). Vacuum bagging setup showing peel ply, flow media, and resin inlet/outlet arrangements. Accessed: 21 April 2026.

After dry lay-up, the laminate pack is ready for vacuum bagging. Layers like peel ply and flow mesh are used, which allow smooth flow of resin and surface preparation after curing. Lines for resin inlet and vacuum outlet are properly arranged so that resin is uniformly distributed throughout the panel. Finally, everything is enclosed in a vacuum bag, which provides an airtight environment. Sealing the components inside the vacuum bag properly is very important, as any air leakages may result in formation of voids.

Stage 3: Vacuum Application and Fiber Compaction.

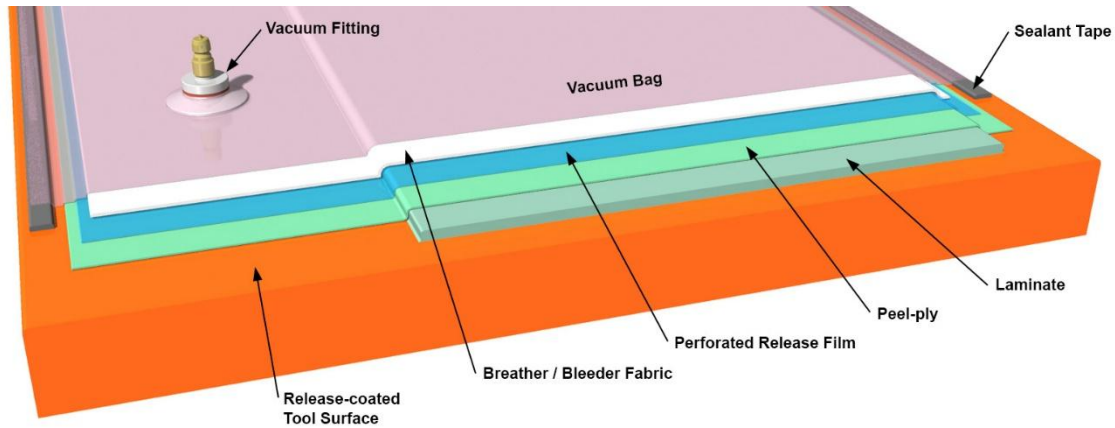


Figure 3.12: Vacuum application and fibre compaction (source: Explore Composites (2022)). Application of vacuum pressure to compact the composite laminate and remove entrapped air prior to resin infusion. Accessed: 21 April 2026.

After the vacuum bagging process is completed, vacuum pressure is used to extract air trapped between the layers of the laminate. This results in the compression of fibres and core material against the mold surface, thus enhancing the volume fraction of fibers and maintaining uniformity in laminate thickness. Vacuum pressure also provides the necessary pressure differential for resin infusion.

Stage 4: Resin infusion and Curing .

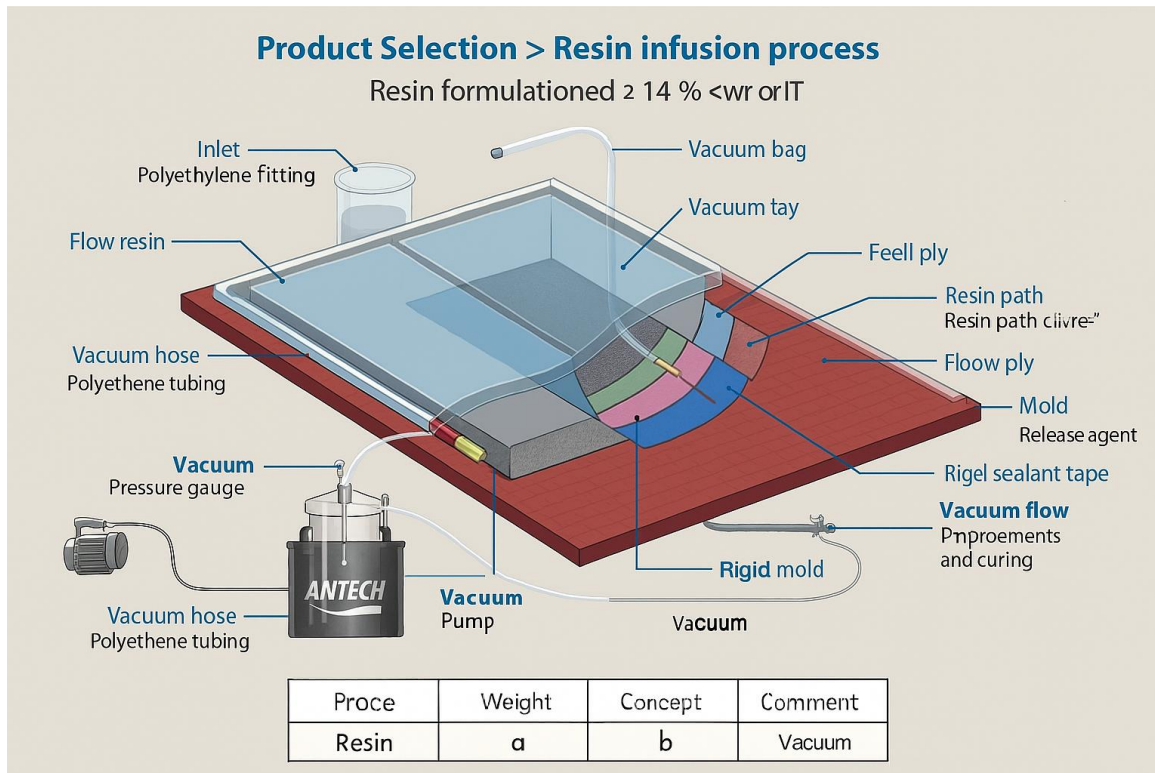


Figure 3.13: Resin infusion and curing stage of the VARI process (source: Mallick (2007)). Epoxy resin flowing through the dry fibre reinforcement under vacuum, followed by curing to form the composite panel. Accessed: 21 April 2026.

In the last phase, epoxy resin is fed via the inlet line and pulled through the dry reinforcement by vacuum. The resin permeates the laminate and core components, fully wetting out the fibers to form the composite panel. Upon completing the wet-out process, the resin is left to cure at room temperature or at an elevated temperature based on the type of resin used. Following curing, the vacuum bag and its associated materials are stripped away, and the composite panel is trimmed, drilled, and sealed along the edges.

Relevance of VARI to this Project.

The VARI method is especially suitable for the fabrication of large and thin composite sandwich panels, such as those required for the construction of Formula Student bodywork parts. It allows for the creation of lightweight panels with high quality, low voidage, and uniform mechanical properties through student accessible equipment.

4.RESULT AND DISCUSSION.

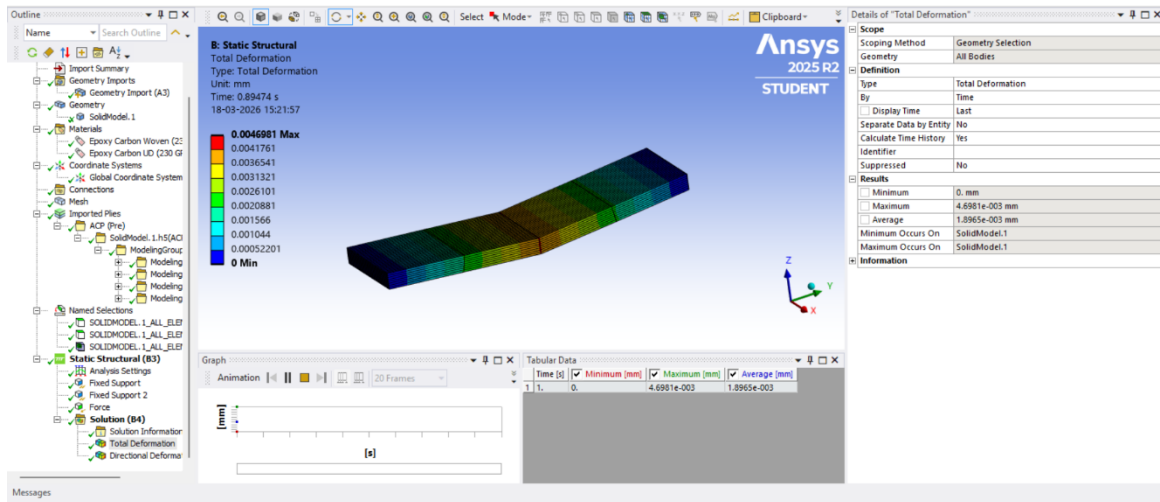


Figure 4.1 Total Deformation Response.

Figure 4.1 depicts the total deformation contour for the CFRP sandwich specimen under the effect of a three-point bending load. The force applied on the mid-span of the 200 mm length panel is equal to 200 N, and there are simple supports at each end of the test specimen. Maximum deformation is noted to take place at the midpoint of the specimen where the force was applied, while deformation progressively decreases towards the edges where there are supports. The total maximum deformation observed in the current test was approximately 4.9×10^{-3} mm, which shows the stiff characteristics of bending exhibited by the CFRP sandwich specimens.

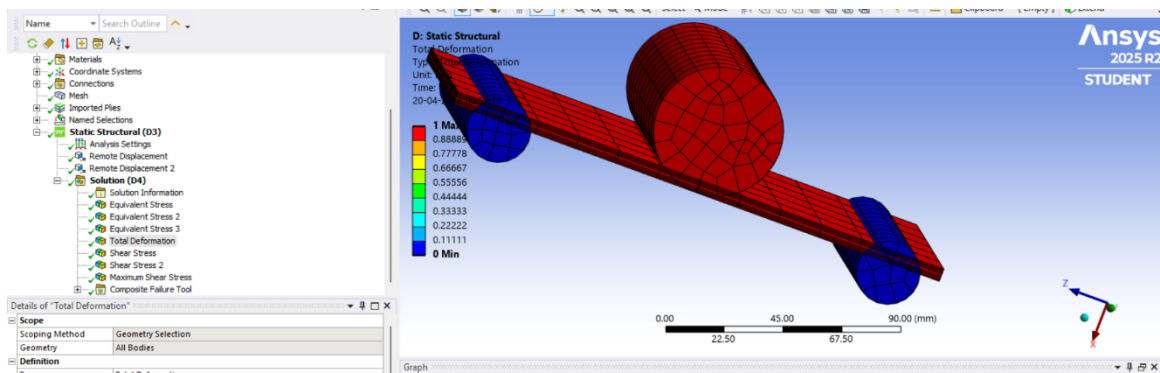


Figure 4.2 Three Point bending test.

Figure above illustrates the geometry of the three-point bending test used in numerical simulation. The composite sandwich panel has a total length of 200 mm, representing the span of the test. Support elements are arranged at each end of the specimen to limit vertical movement but allow rotational movement in order to represent a simply supported situation. A loading force of 200 N acts on the mid-span region of the panel through central loading. This setup will result in bending deformation and stresses being created mainly through flexural action.

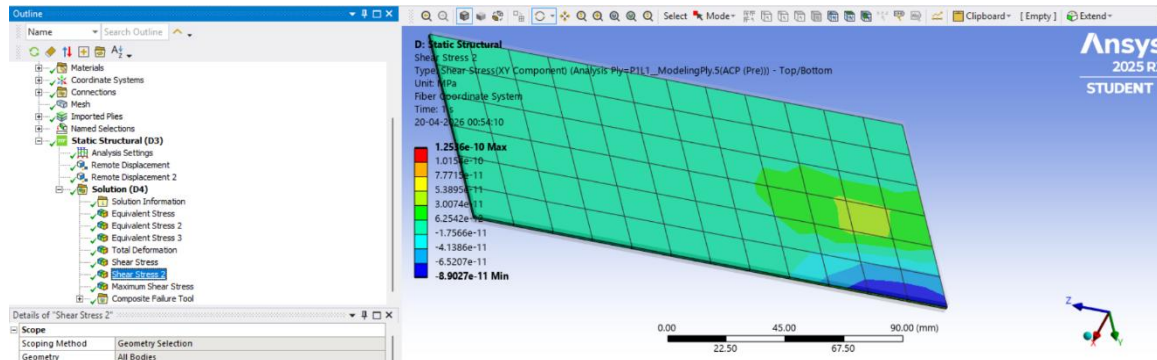


Figure 4.3 Shear Stress Distribution.

Figure above displays the shear stress distribution for the CFRP sandwich panel as predicted by the ANSYS ACP analysis. The shear stresses are localized mainly in regions around the load application point and the support points, since these are regions that experience the greatest transverse forces when subjected to three-point bending. There is relatively low shear stress observed for the rest of the panel length, which indicates that the response of the panel is mainly due to bending, not shearing. Localized regions of shear stresses at the ends of the panels are expected because of the behavior of the sandwich panels.

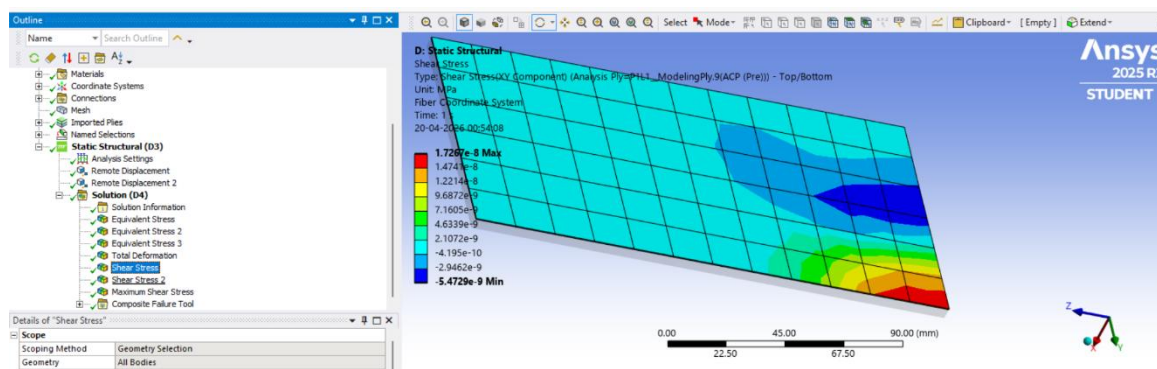


Figure 4.4 Localised Shear Stress Distribution Close to Mid-Span.

Figure 4.3 shows that there is a higher level of shear stress concentration around the point where the load acts centrally. This is due to the transfer of load from the laminate to the core material. The variation in shear stress indicates that there is efficient load transfer

from the composite faces to the PVC foam core. The variation of stress levels also implies that the structure is stable without any discontinuity in the stress levels.

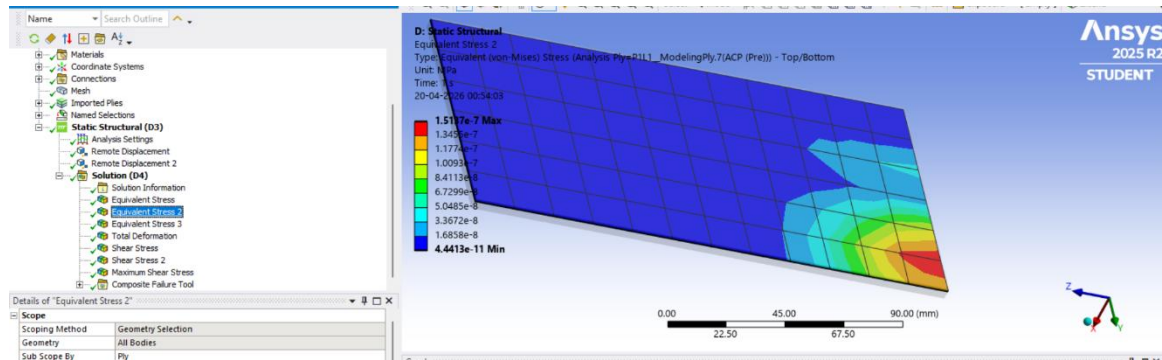


Figure 4.5 Equivalent (Von-Mises) Stress Contours Under Three-Point Bending.

The following diagram shows the stress distribution inside the composite sandwich plate due to the loading effect caused by the three point bending test. Stress is found to be maximum at the loading area and near the support areas with decrease in stress levels with an increase in the distance from these areas towards the midspan. It is evident from the stress distribution pattern that the composite sandwich panel bends through the contribution of face sheets only. While bending takes place, tensile stresses occur in the bottom face sheet and compressive stresses are developed in the top face sheet.

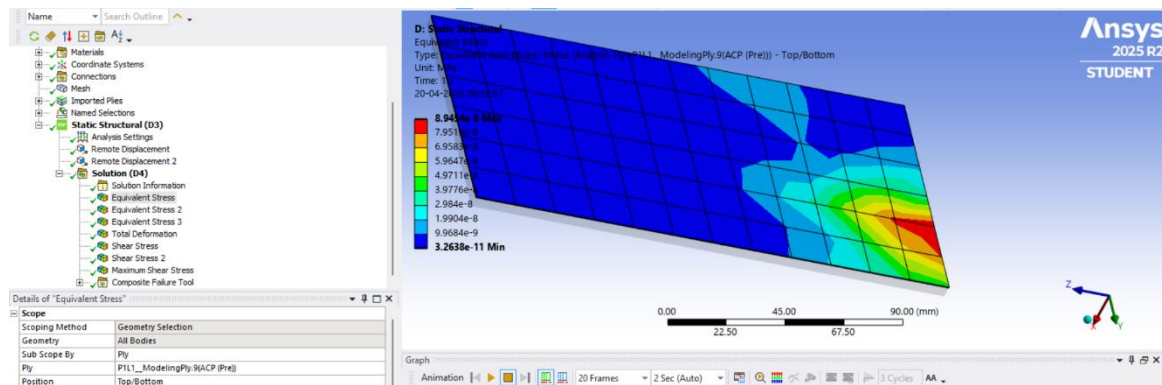


Figure 4.6 Spanwise Variation of Equivalent Stress Across the CFRP Panel.

Figure 4.5 offers another perspective on the distribution of equivalent stresses along the length of the panel to help evaluate the difference in stress level between the mid-span section and the ends. As can be seen, there is a gradual rise in equivalent stresses towards the area of loading and support, which is typical of three-point bending. The fact that there is no peak of stress in other sections of the panel suggests that this geometry and layering help redistribute stresses effectively.

4.1 Integrated Results and Discussion.

The numeric three-point bending analysis performed using ANSYS was used for a detailed study of the behavior of the CFRP sandwich structure subjected to a 200 N load located in the middle of the 200 mm section of the beam. Results from the total deformation analysis showed that the panel demonstrates typical three-point bending behavior, having the maximum total deformation near the center where the load is applied. The maximum total deformation of approximately 4.9×10^{-3} mm showed that the panel had a high level of flexural stiffness, which is typical for the behavior of CFRP sandwich panels. The symmetrical nature of the deformation pattern was confirmed by the directional deformation diagrams that demonstrated that the applied boundary conditions were correct, and only bending was involved.

The distribution of the shear stress demonstrated that these stresses were mainly localized in the core area and in the areas close to the location of the loads and supports. It should be noted that the core was responsible for transferring transverse shear loads in order to separate the two sheets of CFRP. It is important that the gradient of stresses across the whole structure was even; thus, there were no discontinuities.

Equivalent (von-Mises) stress isolines indicated that bending loads were primarily handled by the CFRP face sheets, where the tensile stresses appeared in the bottom face sheet, while the compressive stresses appeared in the top face sheet. The maximum stress values occurred at mid-span and support locations, where the bending moment is high. The analysis based on ACP-based composite material captured the true anisotropic characteristics of the CFRP laminates. This gives better structural performance than the simplified solid approach where the predicted deformations will be lower because of its higher stiffness value. As seen from the simulation results, the overall deformation indicates that the 2 mm thickness CFRP sandwich panel works effectively in the elastic zone. Thus, it can be used efficiently for the body of Formula Student application.

Parameter	Value	Unit	Description
Panel type	CFRP sandwich panel	–	CFRP skins with PVC foam core
Total panel thickness	2	mm	Applied consistently in ACP and solid models
Span length	200	mm	Three-point bending specimen length
Applied load	200	N	Central downward force
Maximum total deformation	4.9×10^{-3}	mm	Occurred at mid-span
Deformation behaviour	Symmetric	–	Confirms correct boundary conditions
Stress concentration	CFRP face sheets	–	Bending-dominated response
Core behaviour	Shear-dominated	–	PVC core transfers shear loads
Structural response	Linear elastic	–	No instability or failure observed
Analysis method	ANSYS (ACP + solid model)	–	Composite-accurate numerical simulation

Table 4.1 Summary of Numerical Three-Point Bending Results.

REFERENCES

REFERENCES

- [1] Formula Student, "Formula Student Autonomous Rules 2026", Institution of Mechanical Engineers, 2025. Available at: <https://www.imeche.org/formulastudent> [Accessed: April 2026]
- [2] Davies, G.A.O., *Materials for Automobile Bodies*, Butterworth-Heinemann, 2003
- [3] Gay, D., Hoa, S.V. and Tsai, S.W., *Composite Materials: Design and Applications*, CRC Press, 2003
- [4] Gibson, R.F., *Principles of Composite Material Mechanics*, 4th ed., McGraw-Hill, 2016
- [5] Hull, D. and Clyne, T.W., *An Introduction to Composite Materials*, Cambridge University Press, 1996
- [6] Mallick, P.K., *Fiber-Reinforced Composites: Materials, Manufacturing, and Design*, 3rd ed., CRC Press, 2007
- [7] Campbell, F.C., *Manufacturing Processes for Advanced Composites*, Elsevier, 2010
- [8] ANSYS Inc., "ANSYS Composite PrepPost (ACP) User Guide", ANSYS Documentation, 2024. Available at: <https://www.ansys.com/documentation> [Accessed: April 2026]
- [9] ASTM International, "ASTM D7264/D7264M – Standard Test Method for Flexural Properties of Polymer Matrix Composite Materials", ASTM International, 2019

BIBLIOGRAPHY

The listed sources have been referred to during the course of the work but have not been cited in the main report text. They provided an overview concerning the areas of composite material science, structural analysis, motorsports, and manufacturing technology as related to the Formula Student concept.

Campbell, F.C. (2004). *Manufacturing Processes for Advanced Composites*. Elsevier Butterworth-Heinemann.

Chung, D.D.L. (2010). *Composite Materials: Science and Applications*. 2nd ed. Springer.

Daniel, I.M. and Ishai, O. (2006). *Engineering Mechanics of Composite Materials*. 2nd ed. Oxford University Press.

Jones, R.M. (1999). *Mechanics of Composite Materials*. Taylor & Francis.

Kaw, A.K. (2006). *Mechanics of Composite Materials*. 2nd ed. CRC Press.

Mazumdar, S.K. (2001). *Composites Manufacturing: Materials, Product, and Process Engineering*. CRC Press.

Strong, A.B. (2008). *Fundamentals of Composites Manufacturing: Materials, Methods and Applications*. Society of Manufacturing Engineers.

Tita, V. (2013). *Failure and Damage Analysis in Advanced Composite Materials*. Springer.

Wiedemann, J. (2015). *Race Car Vehicle Dynamics*. SAE International.

ASTM International (2018). *Standards Related to Composite Materials*. ASTM Publications.

ANSYS Inc. (2023). *Introduction to Composite Materials Modelling*. ANSYS Training Resources.

APPENDICES

Appendix contains supplementary information which supports the content of the report but does not interfere with the technical process. It includes information on project management, numerically modeling information, simulation results, and test setup information. These pieces of information make sure that the appendix doesn't make up the main portion of the report.

APPENDIX A-Project Gantt Chart.

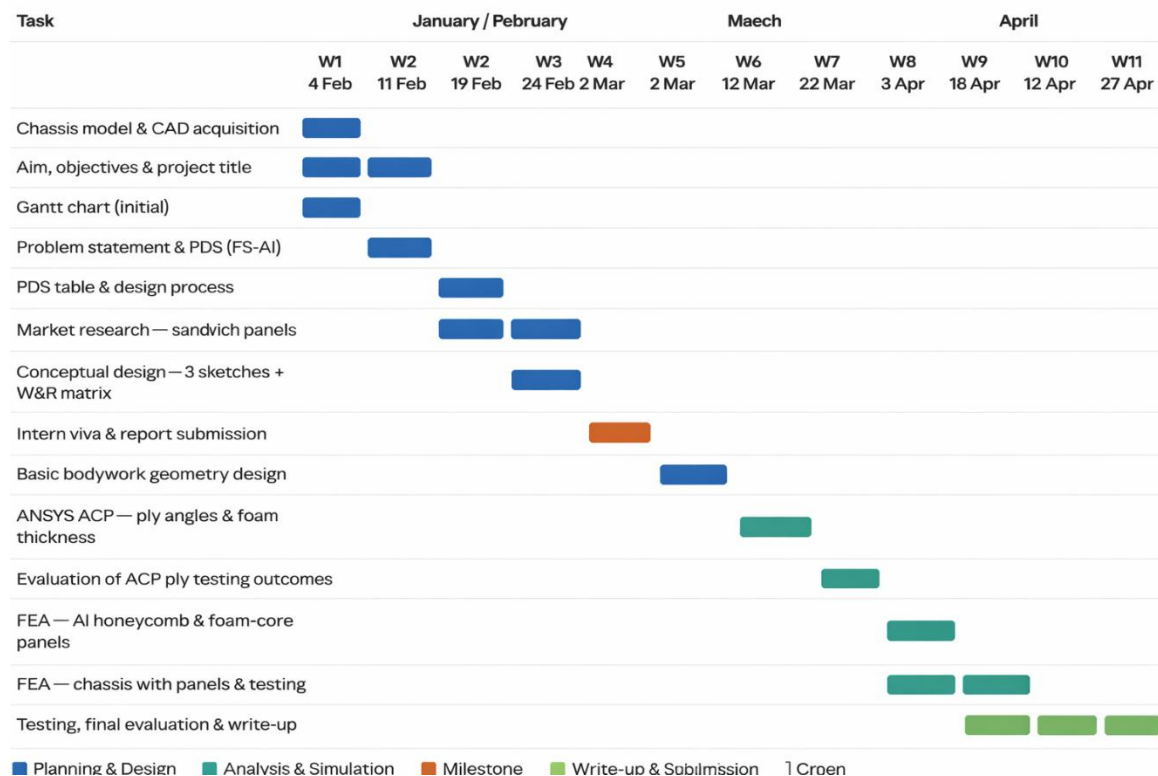


Figure A.1 Gantt chart.

APPENDIX B-ANSYS COMPOSITE PREPOST (ACP).

The screenshots below show the details of the composite laminate definitions made in ANSYS Composite PrepPost (ACP), including the ply stacking sequences, material definitions for the face sheets made of CFRP, core material definition using PVC foam, fibre orientations, and the overall thickness of the composite panel equal to 2 mm. These are based on the composite modelling approach explained in Chapter 3.

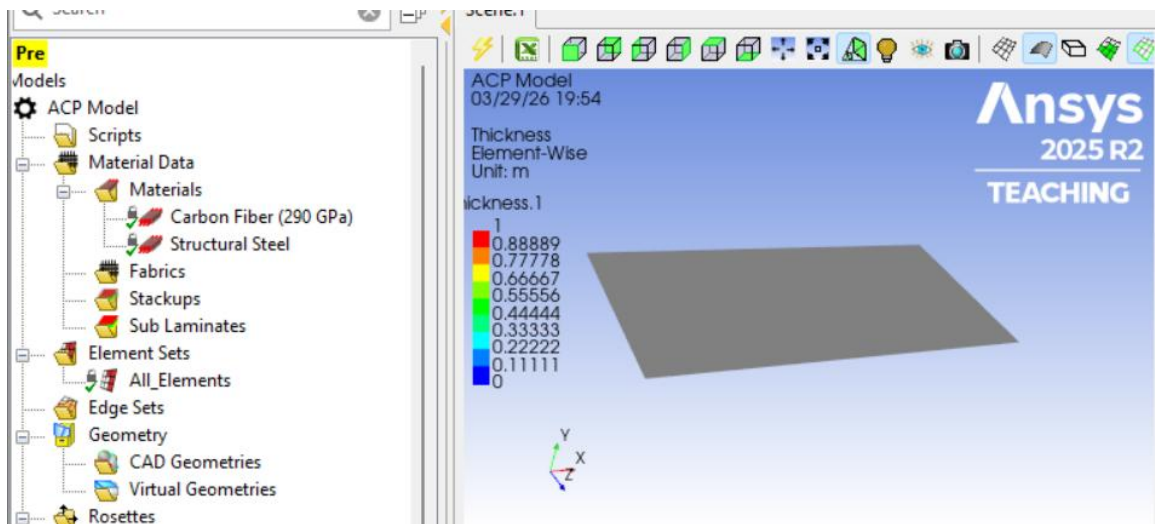


Figure A.2 ANSYS ACP.

APPENDIX C- Additional Numerical Analysis Results.

The following appendix shows some additional numerical data that was derived from the three-point bending analysis, which is not covered in the Results section of the thesis. This includes some additional deformation contour diagrams, shear stress profiles, equivalent stress diagrams, and other visual representations of the reaction of the sandwich panel made of carbon fiber reinforced polymer.

APPENDIX D: LOGBOOK TEMPLATE

Name (SRN):	RUPESHWARAN PARANTHAMAN (23069976)
Project title:	Design & Analysis of Body Work for an Autonomous Racing Vehicle
Supervisor:	ESFANDIAR KHALEGHI

Student notes & date:	
Planned tasks and Work Undertaken:	I aligned the report structure with the School template, completed the market-research summary with references, integrated the full manufacturing methodology (Stages 1–9) with figures, and converted earlier notes into a concise conclusions section suitable for the final report.
Problems encountered/ questions for supervisor:	I require confirmation that an oven post-cure is available and approved, agreement on the edge-radius inspection method (template or flexible radius gauge), and clarity on whether a short CFD screening is required now or listed as future work.

Project Meeting date & staff(s):	
Meeting notes:	04-02-2026-WEEK ONE - Decide on the Chassis Model - Get the CAD of the Chassis & Prev. Body Work - Write Project Aim & Objectives - Gantt Chart
	11-02-2026-WEEK TWO - Aim & Objectives & Project Title - Updated Gantt Chart - Problem Statement - PDS (FS AI) 19-02-2026-WEEK THREE - Graphical Gantt Chart - PDS (Table) - Follow the Design Process - Start Market Research on Sandwich Pannels 24-02-2026-WEEK FOUR - Conceptual Design (3x Sketch with annotations + Pros/Cons) - Concept Evaluation W&R Matrix 02-03-2026- WEEK FIVE - Intern viva and Intern report submission 12-03-2026-week six - Basic design of the geomentry of body work 22-03-2026-WEEK SEVEN - ANSYS ACP material designs, ply angle, foam thickness and testing 08-04-2026-WEEK EIGHT Evaluation of the ansys acp testing outcome for each individual ply

	12-04-2026-WEEK NINE • FEA study of the carbon fibre aluminium honeycomb panel • FEA study of the foam-cored carbon fibre panel 18-04-2026-WEEK TEN • Continuation of the FEA study of the chassis with panels & Test • 27-04-2026-WEEK ELEVEN Testing, final evaluation, and write-up
Action point(s):	I will add the signed academic-integrity page to the front matter, insert the completed Risk and Resources appendices, and prepare a brief QA checklist covering the leak-rate target, resin-temperature window, and edge-sealing procedure.
Checked by:	
Date:	

APPENDIX E: PROJECT RISK AND THREAT ASSESSMENT CHECKLIST

This appendix is to be discussed during the initial supervisory meetings and must be included with the Interim report. Please **PRINT** all information **CLEARLY**

Student's Name:	Rupeshwaran Paranthaman
SRN:	23069976
Project Title:	Design & Analysis of CFRP Bodywork for a Formula Student Autonomous
Supervisor's Name:	Esfandiar Khaleghi

Please enter below your **BEST ESTIMATES** for the information requested.

1	Is your project likely to involve either deliberate, or possible accidental use or contact with: Voltages above 30V High currents at low voltage (i.e. lead acid batteries) Rotating machines High temperatures Hazardous fluids or gases		No
2	Will your project involve attaching electrodes to yourself or someone else in such a way as to produce low impedance contact?		No
3	If the answer to 2 is Yes, has an assessment of the risk in the planned procedure been carried out?	Yes	
4	If the answer to 2 or 3 is Yes, does the analysis show an acceptable level of risk?		No
5	Are any substances classified as being 'Hazardous to Health' likely to be used in the course of your work?	Yes	
6	Are any flammable substances likely to be used in the course of your work? (e.g. volatile cleaning agents, paint, etc.)	Yes	
7	Are there any other aspects of your projected work which might impose a danger to yourself or to others? (Please specify):	Yes	

Notes: If the answer to any of these questions is YES, you have to complete the Risk and Hazard table on the next page and ask your Supervisor to sign and approve it prior to submission of your Project Outline.

APPENDIX F: PROJECT RESOURCES CHECKLIST

This appendix is to be discussed during the initial supervisory meetings and must be included with the Interim report.

Please **PRINT** all information **CLEARLY**

Student's Name:	Rupeshwaran Paranthaman
SRN:	23069976
Project Title:	: Design & Analysis of CFRP Bodywork for a Formula Student Autonomous Vehicle
Supervisor's Name:	Esfandiar Khaleghi
Date:	27/04/2026

Please enter below your **BEST ESTIMATES** for the information requested. You will not be penalised for small errors and later genuine changes - only for not bothering!

1	Will materials or components be ordered?	Yes	
2	Will you be designing a PCB or drawing which will require manufacture during the course of your project? (Note that this should only be contemplated where absolutely necessary for the success of the project!)		No
3	Will PCB or drawing assemblies already designed and manufactured within the School be required? Please specify:		No
4	Will a PC be required for work other than for report writing?	Yes	
5	If the answer to 4 is Yes, will you require any special hardware installed/attached (i.e.PROM emulation, USB interface). Please specify:		No

6	Will you require specific software, other than word processing? (Please specify):	Yes	
7	Will your project require technical staff to make items requiring workshop facilities?	Yes	

8	If the answer to 7 is Yes, which of the following workshop activities will be required (please circle): Drilling Sheet metal forming Lathe work Milling , Other (Please specify)		
9	Are there other resources required, not covered above? (Please specify):	Yes	